



REPUBLIC OF GHANA  
OFFICE OF THE HEAD OF THE CIVIL SERVICE  
INTERNAL AUDIT DEPARTMENT

# Product Requirements Document

IAD Digital Platform — the official web presence and operational backbone of internal audit across the Ghana Civil Service

|              |                                                                                          |
|--------------|------------------------------------------------------------------------------------------|
| DOCUMENT     | Product Requirements Document (PRD)                                                      |
| PRODUCT      | IAD Digital Platform — <a href="https://iad.ohcsghana.org">https://iad.ohcsghana.org</a> |
| PREPARED BY  | OHCS IT Department                                                                       |
| PREPARED FOR | Internal Audit Department (IAD)                                                          |
| VERSION      | 1.5 — issued for submission to IAD management                                            |
| STATUS       | Issued for submission to IAD management                                                  |
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# Document Control

|                   |                                                                                                    |
|-------------------|----------------------------------------------------------------------------------------------------|
| Title             | Product Requirements Document — IAD Digital Platform                                               |
| Version           | 1.5                                                                                                |
| Classification    | Official — Internal (OHCS / IAD)                                                                   |
| Author            | OHCS IT Department                                                                                 |
| Owner             | Director of Internal Audit                                                                         |
| Live platform     | <a href="https://iad.ohcsghana.org">https://iad.ohcsghana.org</a>                                  |
| Source repository | <a href="https://github.com/ghwmlite-dotcom/IAD">github.com/ghwmlite-dotcom/IAD</a> (branch: main) |

## Revision History

| VERSION | DATE     | CHANGE                                                                     |
|---------|----------|----------------------------------------------------------------------------|
| 1.0     | May 2026 | First issue — platform as deployed through Tier 2 (Knowledge Hub)          |
| 1.1     | May 2026 | Annex A: vendor valuation replaced with internal operationalization budget |
| 1.2     | May 2026 | Annex A: budget reduced to GHS 35,000 via free tiers and absorption        |
| 1.3     | May 2026 | Annex A: budget restricted to technical line items only                    |
| 1.4     | May 2026 | Annex A: optional AI module added (GHS 50,000 total if adopted)            |
| 1.5     | May 2026 | Closing note reframed for submission to IAD management                     |

## Approval

|                                                           |                                                  |                                              |
|-----------------------------------------------------------|--------------------------------------------------|----------------------------------------------|
| <b>Director of Internal Audit</b><br>Product Owner — Date | <b>Director, RSIMD</b><br>Delivering Unit — Date | <b>Head of Civil Service</b><br>Noted — Date |
|-----------------------------------------------------------|--------------------------------------------------|----------------------------------------------|

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## 1 Executive Summary

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The IAD Digital Platform is the official web presence and operational backbone of the Internal Audit Department of Ghana's Civil Service. It serves two missions at once:

1. **Public mission** — position the IAD as a transparent, modern, globally credible audit institution: publishing audit findings, the state of internal audit across the service, and a verifiable professional registry of Internal Audit Class officers.
2. **Operational mission** — give the IAD and every MDA Internal Audit Unit a single digital headquarters: audit universe management, annual planning, engagement tracking, findings & recommendations follow-up, and an authoritative Knowledge Hub.

**Institutional context.** The IAD is the headquarters of internal audit in the Ghana Civil Service: every Ministry, Department and Agency (MDA) has its own Internal Audit Unit with its own head, and all of them answer to the Director of Internal Audit. The platform is architected around that reality — it is built for many units, not one department.

## 2 Background & Problem Statement

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Before this platform:

- The IAD had no dedicated digital identity; it existed as a sub-section of the OHCS website with no operational tooling.
- Audit findings and recommendations lived in spreadsheets and paper files; there was no public accountability surface and no consolidated view of implementation status.
- Internal Audit Class officers had no verifiable professional identity — credentials (FCCA, ACCA, IIA, CITG) could not be checked by employers, donors or the public.

Officers' qualifications lived on paper CVs and in personnel files: an employer posting or promoting an officer, a development partner funding a PFM programme and seeking assurance that the audit units involved were competently staffed, or a journalist, researcher or citizen — none had an independent way to confirm that an officer held what they claimed.

- Standards, manuals and templates circulated informally; MDA units had no single authoritative source for IAD guidance.
- Whistleblowing channels lacked provable anonymity guarantees.

### 3 Goals & Success Metrics

| GOAL                     | METRIC                                                                                |
|--------------------------|---------------------------------------------------------------------------------------|
| Transparency leadership  | Press/academic citations of the public findings tracker; donor references             |
| Operational adoption     | MDA liaison accounts active; findings with management responses on file               |
| Professional registry    | Officers with verified digital profiles; certificate verifications served             |
| Knowledge centralisation | Knowledge Hub documents published; downloads served                                   |
| Platform excellence      | Sub-second global TTFB; 95+ Lighthouse accessibility; zero critical security findings |

### 4 Personas & Access Model

| PERSONA                                         | DESCRIPTION                                                                                          | ACCESS                           |
|-------------------------------------------------|------------------------------------------------------------------------------------------------------|----------------------------------|
| <b>Public citizen / journalist / researcher</b> | Consumes transparency data, verifies auditors, downloads publications, submits whistleblower reports | Public pages, no login           |
| <b>MDA audit unit staff (liaison)</b>           | Manages their MDA's audit universe, plans, engagements and findings                                  | Portal — role: liaison           |
| <b>IAD auditor / manager</b>                    | Cross-MDA engagement work, findings review, working papers                                           | Portal — roles: auditor, manager |
| <b>Director of Internal Audit</b>               | Full oversight, publishing decisions, administrative functions                                       | Portal + Admin — role: director  |
| <b>Platform administrator</b>                   | Users, registry, content, Knowledge Hub, site configuration                                          | Admin — role: admin              |

**Authentication.** Passwordless magic-link email sign-in with role-based sessions — a signed-in device is remembered for 30 days (sliding renewal, 30-day hard cap) before re-authentication is required.

## 5 Shipped Scope (as of v1.0)

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### 5.1 Public Website

- Fully bespoke IAD brand and design system: hero slider designed around the internal-audit mission; leadership section featuring the Director of Internal Audit and the OHCS Internal Audit Unit head; news, events, services, contact.
- Bilingual chrome (EN/FR); body content in English.
- SEO/AEO pass: prerendered content pages, JSON-LD structured data, sitemap, and paginated listing pages built to scale to hundreds of entries.
- Mobile-responsive across the entire platform (audited and fixed); newsletter subscription.

### 5.2 Public Findings Tracker (/track)

Public, searchable tracker of audit findings and recommendations with implementation status — the transparency flagship.

### 5.3 "State of Internal Audit" Transparency Page (/transparency)

Live dashboard: audit-plan coverage, aggregate risk heat map, resolution-rate trends. Aggregate queries only, cached at the edge.

### 5.4 Internal Audit Class Registry (/registry, /verify)

Verified professional profiles for Internal Audit Class officers (credentials, CPD), QR-verifiable digital certificates, and a public verification endpoint. Paginated and prerendered for search visibility at scale.

### 5.5 Audit Operations Portal (/portal)

Audit universe (MDAs, auditable entities), risk register, annual planning, engagements with working-paper uploads, findings & recommendations tracker, notifications, and per-MDA liaison views.

### 5.6 Knowledge Hub Tier 2 — shipped

- **Public** (/publications): searchable, paginated library of manuals, templates, standards, circulars, guidelines, reports, forms and policies with versioned, downloadable files.
- **Portal** (/portal/knowledge): adds MDA-only restricted documents.
- **Admin**: full publishing workflow — upload (PDF/DOCX/XLSX ≤ 25 MB), metadata, draft/publish, versioning with change notes, audience control.

### 5.7 Admin Console (/admin)

Registry management, users & roles, Knowledge Hub publishing, submissions (whistleblower/feedback) review pipeline, leadership, news & events, site configuration, and an audit log of every administrative mutation. AI-training features disabled by directive.

### 5.8 Whistleblowing & Submissions

Public reporting form with sealed reference numbers and a review pipeline in admin. Provable-anonymity hardening is scheduled in Phase 3 (see §9).

## 6 Architecture

| LAYER        | TECHNOLOGY                                                               |
|--------------|--------------------------------------------------------------------------|
| Frontend     | Next.js 16 (static export), React 19, Tailwind CSS v4                    |
| Edge hosting | Cloudflare Pages — custom domain iad.ohcsghana.org with SSL              |
| Backend      | Cloudflare Pages Functions (serverless, same-origin /api/*)              |
| Database     | Cloudflare D1 (SQLite at the edge) — 15 migrations applied               |
| File storage | Cloudflare R2 — working papers and Knowledge Hub documents               |
| Auth         | Magic-link email, role-based sessions                                    |
| Quality      | 423 automated tests (Vitest), static build gate, documented API contract |

### Key architectural decisions

- **Static export + edge functions** — the whole site is served from Cloudflare's edge network for sub-second global loads; all dynamic behaviour goes through same-origin APIs.
- **Contract-first backend** — docs/API-CONTRACT.md defines the entire API surface.
- **Audit-everything** — every administrative mutation writes to an audit log.
- **Fail-soft prerendering** — build-time data snapshots degrade gracefully, so builds never break on API unavailability.

## 7 Non-Functional Requirements

- **Performance** — sub-second TTFB worldwide via the edge network; paginated lists (12/page) so pages stay fast as data grows.
- **Security** — role-gated endpoints, session hardening, magic-byte file-type validation on uploads, Content-Disposition sanitisation, no client-side secrets.
- **Privacy** — whistleblower submissions designed for anonymity; Phase 3 adds formal anonymity hardening and a published protocol.
- **Reliability** — idempotent seed/migration tooling; demonstration data clearly separated from production roster data.
- **Accessibility** — semantic HTML, ARIA labelling, keyboard-operable modals and toggles; formal WCAG pass scheduled in Phase 3.

## 8 Known Limitations (v1.0)

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1. Magic-link login requires the production email-service (Resend) API key to be set as a deployment secret.
2. Seeded demonstration data (MDAs, findings, registry officers) awaits replacement with the real roster — bulk CSV import is available as a follow-on.
3. News/events admin screens exist; final wiring of those content types is optional polish.
4. French translation covers site chrome; full body-content translation is Tier 3.

## 9 Roadmap

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| PHASE         | SCOPE                                                                                                                         | STATUS  |
|---------------|-------------------------------------------------------------------------------------------------------------------------------|---------|
| Phase 1       | Live backend: D1 schema, API, magic-link auth, admin CMS                                                                      | Shipped |
| Phase 2       | Audit Operations Platform — universe, risk, planning, engagements, findings, liaison portal                                   | Shipped |
| Tier 1        | Public findings tracker, transparency page, IAC registry & CPD                                                                | Shipped |
| Tier 2        | Knowledge Hub                                                                                                                 | Shipped |
| Phase 3       | Hardening: security review, whistleblower provable-anonymity certification, accessibility pass, performance, launch checklist | Next    |
| Tier 2 (rem.) | AI-assisted audit copilot (internal; gated on security review and real data)                                                  | Planned |
| Tier 3        | Full bilingual content, annual "State of Accountability" microsite, performance & accessibility as brand                      | Planned |

## 10 Risks & Dependencies

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- **Data ownership** — the real officer roster and findings data must come from IAD; the platform's value compounds only when real data flows.
- **Email deliverability** — magic-link authentication depends on the email service key and DNS records remaining valid.
- **Continuity** — repository, hosting account and database live under the OHCS Google account; credentials handover should be formally documented.

## A Annex A — Operationalization Budget

**Internal delivery.** The platform is developed in-house by the OHCS IT Department for the Internal Audit Department. No external development fees apply. The budget below is restricted to **technical line items only** — the engineering work and running costs required to operationalize and keep the platform — so that every figure is directly traceable to the platform itself. Training, launch and documentation activities are absorbed internally by OHCS IT and carry no cost here.

### A. One-off technical costs

| #                  | ITEM                                   | WHAT IT COVERS                                                                                                                                             | GHS    |
|--------------------|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 1                  | Cloud infrastructure provisioning      | Cloudflare Pages/Functions environments, D1 database setup, R2 storage, DNS & SSL configuration, deployment secrets                                        | 6,000  |
| 2                  | Data migration & digitisation          | Engineering work to move the real Internal Audit Class officer roster and findings & recommendations records into the platform database, with verification | 8,000  |
| 3                  | Independent security assessment        | External penetration test of the platform and anonymity review of the whistleblowing channel before go-live                                                | 9,000  |
| 4                  | Backup, monitoring & disaster recovery | Automated database backups, uptime monitoring, alerting and recovery runbooks                                                                              | 4,000  |
| 5                  | Email authentication infrastructure    | Magic-link email service integration and deliverability configuration (SPF/DKIM)                                                                           | 3,000  |
| 6                  | Contingency ( $\approx 10\%$ )         |                                                                                                                                                            | 2,600  |
| Subtotal — one-off |                                        |                                                                                                                                                            | 32,600 |

### B. Recurrent annual costs

| #                    | ITEM                        | WHAT IT COVERS                                | GHS/YR |
|----------------------|-----------------------------|-----------------------------------------------|--------|
| 1                    | Transactional email service | Magic-link authentication emails (entry tier) | 1,800  |
| 2                    | Domain & DNS                | ohcsghana.org renewal share                   | 600    |
| Subtotal — recurrent |                             |                                               | 2,400  |

Cost absorptions that make the GHS 35,000 possible: Cloudflare Pages, D1 and R2 run on **free tiers** at launch volumes (GHS 0); platform maintenance, training, launch and documentation are **absorbed by OHCS IT** as part of normal duties (GHS 0).

D. Optional enhancement — AI-Assisted Audit Capabilities (not mandatory)

**Entirely optional.** This module may be adopted at any later stage without rework — the platform's architecture already anticipates it (see §9, Tier 2: AI-Assisted Audit Copilot). It is presented separately so the core GHS 35,000 ask stands on its own.

| #                             | ITEM                            | WHAT IT COVERS                                                                                                | GHS    |
|-------------------------------|---------------------------------|---------------------------------------------------------------------------------------------------------------|--------|
| 1                             | AI copilot integration          | Cloudflare Workers AI binding, working-paper drafting assistance and sampling support inside the audit portal | 8,000  |
| 2                             | Anomaly-flagging pipeline       | Upload-and-analyse flow for financial data, flagging outliers for auditor review                              | 4,000  |
| 3                             | AI governance & security review | Data-confidentiality controls, prompt isolation, and audit logging of all AI-assisted actions                 | 3,000  |
| Subtotal — optional AI module |                                 |                                                                                                               | 15,000 |

Recurrent impact: negligible at launch — inference runs on the Workers AI free allocation; usage-based costs are reviewed annually.

E. Summary

|                                          | GHS        |
|------------------------------------------|------------|
| Core platform — year one (mandatory)     | 35,000     |
| Optional AI module (one-off, if adopted) | 15,000     |
| Project total with AI module             | 50,000     |
| Subsequent years (core platform)         | 2,400 / yr |

**Conclusion.** The year-one operationalization budget of **GHS 35,000** for the IAD Digital Platform — with an optional AI-assisted audit enhancement of GHS 15,000, taking the total to GHS 50,000 only if adopted — is respectfully submitted to IAD management for consideration and approval. Every line is a technical cost attached to the platform: infrastructure, data, security, resilience and email. All non-technical activities are absorbed in-house by the OHCS IT Department.

